13 Here are two similar solid shapes.

A



B



surface area of shape **A** : surface area of shape **B** = 3 : 4

The volume of shape **B** is 10 cm^3

Work out the volume of shape **A**.
Give your answer correct to 3 significant figures.

Compare A with B. Area scale factor = $\frac{3}{4}$

Length scale factor = $\sqrt{(3 \div 4)} = \frac{\sqrt{3}}{2}$

1 mark
Take the square root of the area factor

Volume scale factor = $(\sqrt{3} \div 2)^3$.

Volume of A = $10 \times (\sqrt{3} \div 2)^3 = 6.495190528\ldots$ cm^3

1 mark
Cube the length factor and multiply the volume of B

1 mark
Round to 3 significant figures

6.50 cm^3

(Total for Question 13 is 3 marks)